

Problem Statement No.8: Document Q&A RAG Agent for Recipe Generator

The Challenge

Food enthusiasts and culinary professionals often rely on unstructured recipe documents, cookbooks, blogs, and user-submitted notes. Searching across these scattered resources is time-consuming, and it's difficult to adapt recipes based on dietary needs, available ingredients, or cultural preferences. The challenge is not just to retrieve recipes but to **intelligently query, adapt, and generate personalized cooking instructions.**

The Objective

Build a **Document Q&A RAG (Retrieval-Augmented Generation) Agent** that can act as an intelligent recipe generator. The solution should:

1. **Ingest & Index Recipe Documents:** Parse cookbooks, PDFs, blogs, or uploaded recipe notes and store them in a searchable knowledge base.
2. **Enable Conversational Q&A:** Allow users to ask natural language questions (e.g., *"How can I make a sugar-free version of chocolate cake?"*) and receive precise, recipe-aware answers.
3. **Create a Personalized & Agentic Layer:** Adapt recipes based on user constraints such as available ingredients, dietary restrictions, cooking time, or cuisine preferences.
4. **Visualize Cooking Guidance:** Present results as step-by-step cooking instructions with substitution suggestions, nutritional facts, and even generate a shopping list.

Tech Stack (Mandatory)

Use **any one** of the following approaches:

- LangFlow + IBM watsonx.ai + Granite Models
- IBM Cloud Agentic Lab with IBM Granite Models
- Replicate with LangChain + RAG + IBM Granite Models